

# Architectural Magnitudes of Feynman Graphs

## A Green-Function Architecture Invariant for Perturbative Quantum Field Theory

### a mathematical contribution introducing a new graph invariant for perturbative Feynman graphs.

Borros Arneth, Philipps University Marburg, Justus Liebig University Giessen, Germany,  
borros.arneth@staff.uni-marburg.de

#### Abstract

Perturbative quantum field theory is formulated in terms of Green functions and Feynman diagrams, which provide one of the most successful computational frameworks in modern theoretical physics. Although Feynman graphs possess well-known topological invariants such as the first Betti number and graph polynomials, no invariant has been proposed that simultaneously characterizes propagation, interaction, holonomy, and recursive structure within a single multiplicative quantity. In this work, we introduce the **Architectural Magnitude** of a Feynman graph, a new graph invariant derived from the four primitive structural sectors of Green-Function Architecture Theory (GFAT). Every finite perturbative Feynman graph is assigned a unique architectural coordinate  $(a, b, c, d)$ , representing propagation, interaction, holonomy, and recursion. From these coordinates we construct the Architectural Magnitude

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8},$$

Where  $A = \left(1 - \frac{1}{3\pi}\right)^{-1}$ . Within the proposed axiomatic framework, the Architectural Magnitude is shown to be a multiplicative graph invariant possessing natural transformation properties under elementary graph operations. Unlike conventional perturbative amplitudes, the proposed invariant depends exclusively on graph architecture and is independent of coupling constants, kinematics, and particle masses. The relationship of the Architectural Magnitude to established graph-theoretical invariants, Green functions, and perturbative quantum field theory is discussed together with first exploratory applications to particle reactions and decay processes. The proposed framework introduces an architectural perspective on Feynman graphs and suggests a new invariant-theoretic approach to perturbative quantum field theory.

**Keywords:** Green functions; Feynman diagrams; graph invariants; perturbative quantum field theory; architectural magnitude; graph theory; topology.

## 1. Introduction

Green functions and Feynman diagrams are among the most fundamental mathematical structures in theoretical physics. Green functions describe the propagation of fields and particles through linear differential operators and constitute the mathematical foundation of quantum mechanics, statistical mechanics, condensed-matter physics, and quantum field theory [1–9]. Feynman diagrams, introduced in 1949, transformed perturbative quantum field theory into a graphical calculus in which propagators, interaction vertices, and loop corrections encode scattering amplitudes in an intuitive and computationally efficient manner [10–16].

Besides their physical interpretation, Feynman diagrams are finite graphs. Consequently, they possess numerous graph-theoretical properties that have become increasingly important in mathematical physics. Examples include the first Betti number, Kirchhoff and Symanzik graph polynomials, and the Hopf-algebraic structure of renormalization developed by Connes and Kreimer [17–23]. These invariants characterize important combinatorial and topological aspects of perturbative quantum field theory.

Despite these developments, existing graph invariants describe only selected structural features of perturbative diagrams. The first Betti number measures the number of independent loops, graph polynomials encode spanning trees and momentum flow, while Hopf algebras describe the recursive combinatorics of renormalization. To our knowledge, no existing invariant simultaneously incorporates propagation, interaction, holonomy, and recursion into a single multiplicative architectural quantity.

The present work proposes such an invariant.

The central idea is to regard every finite perturbative Feynman graph as possessing an underlying **Process Green-Function Architecture**. Rather than treating propagators, interaction vertices, loops, and recursive insertions merely as graphical elements used for perturbative calculations, we interpret them as manifestations of four primitive architectural sectors:

- propagation,
- interaction,
- holonomy,
- recursion.

These four sectors define an architectural coordinate system for perturbative graphs. Every finite Feynman graph is assigned a unique coordinate

$$(a, b, c, d),$$

from which a single multiplicative invariant—the **Architectural Magnitude**—is constructed.

Unlike perturbative amplitudes, the proposed invariant contains no coupling constants, masses, momenta, spinors, polarization vectors, or gauge-dependent quantities. It is intended to characterize only the intrinsic structural organization of the graph itself.

The objectives of this paper are therefore deliberately modest and mathematically focused.

First, we define a graph-theoretical mapping that assigns every finite perturbative Feynman graph a unique architectural coordinate.

Second, we introduce the Architectural Magnitude as a multiplicative invariant associated with these coordinates.

Third, we investigate several elementary mathematical properties of this invariant, including multiplicativity, invariance under graph isomorphisms, and its behavior under elementary graph operations.

Finally, we discuss the relationship between the proposed invariant and existing graph-theoretical constructions in perturbative quantum field theory.

The present work does **not** propose a modification of quantum field theory, the Standard Model, or the Feynman rules. Instead, it introduces an additional mathematical invariant defined on the space of perturbative Feynman graphs. Whether this invariant ultimately carries physical information beyond its graph-theoretical definition remains an open question. Nevertheless, the existence of a natural architectural coordinate system and its associated multiplicative invariant suggests a new mathematical perspective on perturbative quantum field theory that may warrant further investigation.

## 2. Process Green-Function Architectures

The central objective of this work is to associate every finite perturbative Feynman graph with an intrinsic architectural structure. Rather than introducing new dynamical variables or modifying the perturbative rules of quantum field theory, we define an additional mathematical object that characterizes the structural organization of a graph independently of its physical realization.

The basic object of the present theory is the **Process Green-Function Architecture**.

### 2.1 Primitive Architectural Sectors

Perturbative Feynman graphs contain four elementary structural components that appear universally throughout quantum field theory.

These are

- propagation,
- interaction,
- holonomy,
- recursion.

Within conventional perturbation theory these objects arise naturally as propagator lines, interaction vertices, closed loops, and recursive Dyson insertions. In the present framework they are regarded as the primitive architectural sectors of a process.

Definition 1 (Process Green-Function Architecture)

A **Process Green-Function Architecture** is the ordered quadruple

$$\boxed{\mathcal{A} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}),}$$

where

- $\mathcal{P}$  denotes the propagation sector,
- $\mathcal{I}$  denotes the interaction sector,
- $\mathcal{H}$  denotes the holonomy sector,
- $\mathcal{R}$  denotes the recursion sector.

The four sectors are regarded as independent structural components describing the architecture of a perturbative process.

Remark

The Process Green-Function Architecture is not itself a Feynman graph. Rather, it is an abstract structural object associated with a graph. Consequently, different perturbative diagrams may possess identical architectural descriptions.

## 2.2 Architectural Coordinates

To every Process Green-Function Architecture we associate four non-negative integers.

## Definition 2 (Architectural Coordinates)

Let

$$D = (V, E)$$

be a finite connected perturbative Feynman graph.

The **architectural coordinates** of  $D$  are defined by

$$(a, b, c, d) = (N_p, N_l, \beta_1, R),$$

where

- $N_p$  is the number of internal propagators,
- $N_l$  is the number of interaction vertices,
- $\beta_1$  is the first Betti number,
- $R$  denotes the recursion order.

Accordingly,

- $a$  measures propagation,
- $b$  measures interaction,
- $c$  measures holonomy,
- $d$  measures recursion.

## Discussion

The first three coordinates are standard graph-theoretical quantities or are directly determined from the graph.

The fourth coordinate,

$$R,$$

is introduced as an architectural measure of recursive graph construction.

For tree diagrams,

$$R = 0.$$

Graphs generated through recursive Dyson insertions satisfy

$$R > 0.$$

A precise recursive definition of  $R$  will be discussed in Section 4.

**Definition 3 (Graph–Architecture Mapping)**

The mapping

$$\boxed{\Phi: \mathcal{G} \rightarrow \mathbb{N}_0^4}$$

defined by

$$\Phi(D) = (a, b, c, d),$$

is called the **Graph–Architecture Mapping**.

Here

$$\mathcal{G}$$

denotes the class of finite perturbative Feynman graphs.

**Theorem 1 (Existence of Architectural Coordinates)**

Every finite connected perturbative Feynman graph possesses a well-defined architectural coordinate

$$(a, b, c, d).$$

Proof

Let

$$D = (V, E)$$

be a finite connected perturbative Feynman graph.

Since the graph is finite,

the numbers

$$N_P$$

and

$$N_I$$

are finite integers.

The first Betti number

$$\beta_1 = |E| - |V| + 1$$

is a standard graph invariant and is uniquely determined by the graph topology.

Finally,

the perturbative construction uniquely determines the recursive generation history from which the recursion order

$$R$$

is assigned according to Definition 2.

Consequently,

$$(a, b, c, d) = (N_P, N_I, \beta_1, R)$$

exists.

■

### Comment

The existence theorem is elementary but fundamental.

It establishes that every perturbative Feynman graph possesses a canonical position within a four-dimensional architectural coordinate space.

The remainder of the paper investigates the mathematical properties of this coordinate system.

## 2.3 Architectural Equivalence

The architectural coordinates naturally induce an equivalence relation on perturbative graphs.

Definition 4 (Architectural Equivalence)

Two finite perturbative Feynman graphs

$$D_1, D_2$$

are called **architecturally equivalent** if

$$\boxed{\Phi(D_1) = \Phi(D_2)}.$$

Equivalently,

$$(a_1, b_1, c_1, d_1) = (a_2, b_2, c_2, d_2).$$

Architecturally equivalent graphs belong to the same architectural class.

### Theorem 2 (Architectural Equivalence Relation)

Architectural equivalence defines an equivalence relation on the class of finite perturbative Feynman graphs.

Proof

The relation is

### **Reflexive**

since every graph has identical architectural coordinates to itself.

It is

### **Symmetric**

because equality of coordinate tuples is symmetric.

Finally,

it is

### **Transitive**

because equality of tuples is transitive.

Therefore architectural equivalence satisfies the axioms of an equivalence relation.

■

## **2.4 Architectural Space**

The preceding definitions motivate the introduction of an architectural configuration space.

Definition 5 (Architectural Space)

The **architectural space of perturbative Feynman graphs** is defined by

$$\boxed{\mathcal{M}_{\text{arch}} = \mathbb{N}_0^4},$$

whose coordinates are

$$(a, b, c, d).$$

Every finite perturbative Feynman graph determines one point in this space.

Conversely, not every point of

$$\mathcal{M}_{\text{arch}}$$

necessarily corresponds to a realizable perturbative graph.

Determining the admissible region of architectural space constitutes an interesting open problem.

### 3. Architectural Magnitudes

The architectural coordinates introduced in the previous section describe the structural composition of a perturbative Feynman graph. We now seek a single quantity that summarizes this information while preserving the multiplicative character of independent architectural operations.

The desired quantity should satisfy several natural mathematical requirements.

First, it should depend only on the architectural coordinates and therefore be independent of particle masses, coupling constants, momenta, spin, gauge choice, and other dynamical quantities.

Second, it should be invariant under graph isomorphisms.

Third, independent graph compositions should correspond to multiplication of the invariant.

These requirements motivate the following definition.

#### Definition 6 (Architectural Magnitude)

Let

$$D$$

be a finite perturbative Feynman graph with architectural coordinates

$$(a, b, c, d).$$

The **Architectural Magnitude** of  $D$  is defined by

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8},$$

where

$$A = \left(1 - \frac{1}{3\pi}\right)^{-1}.$$

The quantity

$$\Pi(D)$$

is dimensionless.

Remark

The definition depends exclusively upon the architectural coordinates.

Consequently,

$$\Pi(D)$$

contains no information concerning

- particle masses,
- coupling constants,
- external momenta,
- Lorentz indices,
- gauge symmetry,
- spin,
- flavour.

It is therefore proposed as a **pure graph invariant**.

### 3.1 Why a Multiplicative Invariant?

The choice of a multiplicative rather than additive invariant is motivated by the way perturbative graphs are assembled.

Independent architectural contributions accumulate through repeated application of elementary structural operations.

Consequently, the invariant should satisfy

$$\Pi(D_1 \sqcup D_2) = \Pi(D_1)\Pi(D_2).$$

This property is analogous to multiplicative graph invariants encountered elsewhere in combinatorics and mathematical physics.

### **Axiom 1 (Architectural Independence)**

Propagation,

interaction,

holonomy,

and

recursion

contribute independently to the architectural structure of a perturbative graph.

Therefore the total invariant factorizes,

$$\Pi = P(a) I(b) H(c) R(d).$$

### **Axiom 2 (Architectural Homogeneity)**

Each additional primitive operation contributes the same elementary structural increment regardless of where it occurs within the graph.

Hence

$$P(a + 1) = \lambda_P P(a),$$

with analogous relations for

interaction,

holonomy,

and

recursion.

### **Theorem 3 (Exponential Representation)**

Assume Axioms 1 and 2.

Then every multiplicative architectural invariant possesses the representation

$$\boxed{\Pi = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d},$$

for suitable positive constants

$$\lambda_P, \lambda_I, \lambda_H, \lambda_R.$$

Proof

By Axiom 2,

$$P(a + 1) = \lambda_P P(a).$$

Repeated application yields

$$P(a) = \lambda_P^a P(0).$$

Normalizing

$$P(0) = 1$$

gives

$$P(a) = \lambda_P^a.$$

The same argument applies independently to

interaction,

holonomy,

and

recursion.

Therefore

$$\Pi = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d.$$

■

Discussion

Theorem 3 is important because it shows that **the exponential form is not chosen arbitrarily**. Once multiplicativity and homogeneity are assumed, any invariant of this type must have an exponential representation. What remains is to determine the four base constants.

### 3.2 Architectural Normalization

Instead of immediately assigning numerical values to the four bases, we first formulate them abstractly,

$$\Pi = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d.$$

The numerical values

$$2, 3, 3\pi, A^{1/8}$$

will be introduced only after establishing additional architectural principles.

This separation has two advantages.

First, it distinguishes the **general mathematical theory** from one specific realization.

Second, it makes clear that the exponential structure is the primary mathematical result, whereas the choice of bases represents a particular GFAT model.

#### Theorem 4 (Normalization Freedom)

Let

$$\lambda_P, \lambda_I, \lambda_H, \lambda_R > 0.$$

Then every choice of positive bases defines a multiplicative architectural invariant on the coordinate space

$$\mathbb{N}_0^4.$$

Consequently, the problem of constructing an Architectural Magnitude separates naturally into two parts:

1. the general theory of multiplicative architectural invariants;
2. the determination of the physically preferred normalization.

■

#### 4. Canonical Normalization of the Architectural Magnitude

Section 3 established that every multiplicative architectural invariant possesses the general representation

$$\Pi(D) = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d,$$

where the four positive constants

$$\lambda_P, \lambda_I, \lambda_H, \lambda_R$$

remain undetermined.

The present section investigates whether these bases can be derived from internal structural principles rather than introduced by definition.

##### 4.1 Canonical Normalization Problem

The central mathematical problem may be formulated as follows.

Problem 1

Determine whether there exists a unique quadruple

$$(\lambda_P, \lambda_I, \lambda_H, \lambda_R)$$

satisfying the structural axioms of Green-Function Architecture Theory.

Unlike Section 3,

this is now a uniqueness problem.

## 4.2 Structural Principles

We postulate four normalization principles.

Principle I

Propagation possesses the smallest non-trivial connected architecture.

Principle II

Interaction possesses the smallest non-trivial branching architecture.

Principle III

Holonomy possesses the smallest non-trivial closed architecture.

Principle IV

Recursion possesses the smallest self-consistent recursive architecture.

These principles determine only the normalization,  
not the exponential representation itself.

### 4.3 Propagation

Consider a connected graph consisting of one edge.

It possesses exactly

two

boundary points.

No connected transport object possesses fewer.

#### Theorem 5

(Binary Normalization)

If primitive propagation is normalized by the smallest connected transport graph,  
then

$$\boxed{\lambda_p = 2.}$$

Proof

A primitive transport graph consists of one edge joining two vertices.

Hence the primitive propagation unit possesses cardinality

two.

Normalizing one propagation step by this primitive graph yields

$$\lambda_p = 2.$$

■

Remark

Notice that this argument is now **graph-theoretical**, not physical.

The number 2 arises from the combinatorics of connected transport graphs.

#### 4.4 Interaction

The smallest interaction graph contains

three

branches.

Any graph containing only two branches is merely propagation.

#### Theorem 6

(Ternary Interaction Normalization)

If primitive interaction is normalized by the smallest connected branching graph,

then

$$\boxed{\lambda_I = 3.}$$

Proof

The smallest branching graph contains one interaction node together with three adjacent transport channels.

Consequently

interaction possesses primitive degree

three.

Hence

$$\lambda_I = 3.$$

■

Again,

the argument is graph-theoretical.

#### 4.5 Holonomy

Holonomy differs fundamentally from propagation and interaction.

Propagation concerns edges.

Interaction concerns branching.

Holonomy concerns

closed cycles.

The first Betti number

$$\beta_1$$

already counts independent loops.

The question is therefore:

How should one normalize one elementary loop?

Observation

A loop possesses

- closure,
- orientation,
- angular transport.

The simplest architectural closure consists of one complete transport cycle.

At present, GFAT postulates

$$\boxed{\lambda_H = 3\pi.}$$

This choice reflects the triple-sector decomposition introduced previously.

Conjecture 1

Triple-Holonomy Normalization

The primitive holonomy normalization equals

$$3\pi.$$

Discussion

Unlike propagation and interaction,

this normalization is **not yet derived**.

It remains an architectural hypothesis.

A future derivation should relate

$$3\pi$$

to

- Wilson loops,
- geometric holonomy,
- or transport curvature.

## 4.6 Recursion

Similarly,

recursive closure is represented by

$$A = \left(1 - \frac{1}{3\pi}\right)^{-1}.$$

Conjecture 2

Recursive Normalization

Primitive recursion is generated by the smallest fixed-point architecture.

Consequently

$$\lambda_R = A^{1/8}.$$

Again,

this remains conjectural.

#### 4.7 Canonical GFAT Normalization

Combining the proven graph-theoretical normalization results with the architectural hypotheses yields the canonical GFAT realization

$$\boxed{\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}.$$

#### Theorem 7

(Canonical GFAT Magnitude)

Assume

- Binary Propagation,
- Ternary Interaction,
- Triple-Holonomy,
- Recursive Closure.

Then

$$\boxed{\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}.}$$

Proof

Immediate from

Theorems 3–6 together with

Conjectures 1–2.

## 5. Invariant Theory of Feynman Graphs

The previous sections introduced the Architectural Magnitude

$$\Pi(D) = \lambda_P^a \lambda_I^b \lambda_H^c \lambda_R^d,$$

together with its canonical GFAT realization

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}.$$

The purpose of the present section is to investigate its mathematical properties independently of any physical interpretation.

Throughout this section, let

$$\mathcal{G}$$

denote the class of finite connected perturbative Feynman graphs.

### 5.1 Isomorphism Invariance

Any graph invariant must assign identical values to isomorphic graphs.

This property is therefore the first consistency requirement.

Theorem 8 (Graph-Isomorphism Invariance)

Let

$$D_1, D_2 \in \mathcal{G}$$

be graph-isomorphic.

Then

$$\boxed{\Pi(D_1) = \Pi(D_2)}.$$

Proof

Let

$$\varphi: D_1 \rightarrow D_2$$

be a graph isomorphism.

Then

$$\begin{aligned} N_P(D_1) &= N_P(D_2), \\ N_I(D_1) &= N_I(D_2), \end{aligned}$$

and

$$\beta_1(D_1) = \beta_1(D_2),$$

since graph isomorphisms preserve adjacency, incidence, and cycle structure.

If the recursion index  $R$  is defined intrinsically (that is, invariant under graph isomorphism), then

$$R(D_1) = R(D_2).$$

Therefore

$$(a \cdot b \cdot c \cdot d)$$

is identical for both graphs.

Hence

$$\Pi(D_1) = \Pi(D_2).$$

■

---

Remark

This theorem depends critically on **how  $R$  is defined**. If  $R$  depends on a particular perturbative construction history rather than the graph itself, then it is not a graph invariant. A precise intrinsic definition of  $R$  is therefore essential for the theory.

---

## 5.2 Multiplicativity

Graph invariants often possess additive or multiplicative composition laws.

The Architectural Magnitude is designed to be multiplicative.

Theorem 9 (Multiplicativity)

Let

$$D = D_1 \sqcup D_2$$

be the disjoint union of two perturbative graphs.

Then

$$\boxed{\Pi(D) = \Pi(D_1)\Pi(D_2).}$$

Proof

The architectural coordinates satisfy

$$\begin{aligned} a &= a_1 + a_2, \\ b &= b_1 + b_2, \end{aligned}$$

$$c = c_1 + c_2.$$

Assuming the recursion coordinate is additive,

$$d = d_1 + d_2.$$

Therefore

$$\Pi(D) = \lambda_P^{a_1+a_2} \lambda_I^{b_1+b_2} \lambda_H^{c_1+c_2} \lambda_R^{d_1+d_2},$$

which factorizes into

$$\Pi(D_1)\Pi(D_2).$$

■

Remark

Again, the additivity of  $d$  is an assumption that should eventually follow from a rigorous definition of recursion order.

### 5.3 Elementary Graph Operations

Instead of viewing  $\Pi$  as a static invariant, we now study how it transforms under elementary graph operations.

This is one of the principal motivations for introducing the Architectural Magnitude.

Definition 7 (Elementary Architectural Operations)

The following elementary graph operations are distinguished:

1. insertion of one propagator,
2. insertion of one interaction vertex,
3. insertion of one independent loop,
4. recursive graph insertion.

Each operation changes exactly one architectural coordinate.

Theorem 10 (Propagation Transformation)

Insertion of one propagator changes the Architectural Magnitude according to

$$\boxed{\Pi \mapsto \lambda_P \Pi.}$$

In the canonical GFAT normalization,

$$\boxed{\Pi \mapsto 2\Pi.}$$

Proof

Insertion of one propagator increases

$$a$$

by one while leaving

$$b, c, d$$

unchanged.

Therefore

$$\Pi' = \lambda_P^{a+1} \lambda_I^b \lambda_H^c \lambda_R^d = \lambda_P \Pi.$$

■

Theorem 11 (Interaction Transformation)

Insertion of one interaction vertex satisfies

$$\boxed{\Pi \mapsto \lambda_I \Pi.}$$

In the canonical normalization,

$$\boxed{\Pi \mapsto 3\Pi.}$$

Proof

Identical to Theorem 10.

■

Theorem 12 (Holonomy Transformation)

Insertion of one independent loop satisfies

$$\boxed{\Pi \mapsto \lambda_H \Pi.}$$

For the canonical GFAT realization,

$$\boxed{\boxed{\Pi \mapsto 3\pi \Pi.}}$$

Theorem 13 (Recursion Transformation)

Insertion of one elementary recursive operation satisfies

$$\boxed{\Pi \mapsto \lambda_R \Pi.}$$

or

$$\boxed{\Pi \mapsto A^{1/8} \Pi}$$

under the canonical normalization.

## 5.4 Transformation Algebra

The four elementary operations generate a commutative multiplicative algebra acting on the set of Architectural Magnitudes.

Indeed,

$$T_P T_I = T_I T_P,$$

$$T_P T_H = T_H T_P,$$

etc.,

because each operation modifies a different architectural coordinate.

Consequently,

the transformation operators satisfy

$$T_i T_j = T_j T_i,$$

for all primitive sectors.

This immediately implies

Theorem 14 (Commutativity)

The elementary architectural transformations commute.

Discussion

This theorem is mathematically interesting because it shows that the four primitive sectors are independent generators of the invariant. The transformation algebra is therefore isomorphic to the free commutative monoid generated by the four elementary operations.

## 6. Relation to Perturbative Quantum Field Theory

The Architectural Magnitude introduced in the previous sections is intended as an additional graph invariant associated with perturbative Feynman graphs.

The present section investigates its relationship to established mathematical structures of perturbative quantum field theory.

The objective is not to replace the conventional formalism but to determine whether the proposed invariant complements existing graph-theoretical descriptions.

### 6.1 Feynman Graphs as Mathematical Objects

Perturbative quantum field theory associates every process with a finite graph

$$D = (V, E),$$

whose vertices represent interaction events and whose edges represent propagators.

To every graph one assigns a Feynman amplitude

$$\mathcal{M}(D),$$

constructed from the Feynman rules of the underlying quantum field theory [6–10].

The graph itself, however, possesses a rich mathematical structure that exists independently of the amplitude.

## 6.2 Existing Graph Invariants

Several important invariants are already associated with perturbative graphs.

### (i) First Betti Number

The quantity

$$\beta_1 = |E| - |V| + 1$$

counts the number of independent loops.

It measures only one aspect of graph topology.

### (ii) Kirchhoff Polynomial

The first Symanzik polynomial

encodes spanning trees.

It plays a central role in parametric Feynman integrals.

## (iii) Second Symanzik Polynomial

The second graph polynomial

encodes

- external momenta

and

- masses.

## (iv) Hopf Algebra

The Connes–Kreimer Hopf algebra describes

recursive subdivergences,

renormalization,

and graph insertions.

**6.3 Position of the Architectural Magnitude**

The Architectural Magnitude differs fundamentally from the previous constructions.

Unlike

the Betti number,

it does not measure only loops.

Unlike

Symanzik polynomials,

it does not depend on external kinematics.

Unlike

Hopf algebras,

it is not restricted to recursive subdivergences.

Instead,

it combines four independent architectural sectors into one multiplicative quantity.

Consequently,

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}$$

may be interpreted as an architectural graph invariant.

Important Observation

Notice that

the first Betti number

already appears inside

$\Pi$ .

Thus

GFAT contains

the Betti number

as one component,

rather than replacing it.

## 6.4 Independence from the Feynman Rules

The Architectural Magnitude is remarkable because it can be computed

before

any Feynman rules are applied.

Indeed,

one requires only

- the graph,
- the architectural coordinates.

No knowledge of

- coupling constants,
- gauge symmetry,
- particle masses,
- momenta,
- spin,

is required.

Thus

$$\Pi(D)$$

is a purely combinatorial quantity.

Theorem 15

(Graph-Theoretical Independence)

The Architectural Magnitude is independent of the analytical evaluation of the corresponding Feynman amplitude.

Proof

The definition

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}$$

depends exclusively upon

$$(a, b, c, d).$$

These coordinates are determined from the graph itself.

Since neither

the propagator functions,

nor

vertex tensors,

nor

coupling constants,

nor

external kinematics

enter the definition,

the value of

$$\Pi(D)$$

is independent of the analytical evaluation of the perturbative amplitude.

■

## 6.5 Architectural Magnitude as a Universal Graph Invariant

The preceding theorem suggests an important interpretation.

The Architectural Magnitude belongs to the graph,

not to the quantum field theory.

Different quantum field theories

(QED,

QCD,

electroweak theory)

may generate identical perturbative graphs.

These graphs therefore possess identical

architectural magnitudes,

even though

their amplitudes differ.

Consequently,

$$\Pi(D)$$

appears to be a universal invariant of perturbative graph architecture.

### **Conjecture 3 (Universality)**

For every perturbative quantum field theory,

the Architectural Magnitude depends only on the graph topology and architectural structure,

not on the particular interaction described by the theory.

## **7. First Physical Applications of the Architectural Magnitude**

The previous sections established the Architectural Magnitude as a multiplicative invariant of finite perturbative Feynman graphs. The present section explores whether this invariant may contain information relevant to experimentally observable quantities.

The objective is exploratory rather than conclusive. We investigate whether the Architectural Magnitude exhibits systematic relationships with particle masses, reaction energies, and transition probabilities.

### **7.1 Static and Dynamic Architectures**

A central distinction of Green-Function Architecture Theory is between **static** and **dynamic** architectures.

### Definition 8 (Static Architecture)

The static architecture of a particle is the architectural structure associated with an isolated particle state.

Its Architectural Magnitude is

$$\Pi_{\text{particle}}.$$

This quantity characterizes the intrinsic structural complexity of the particle.

### Definition 9 (Dynamic Architecture)

The dynamic architecture of a process is the architectural structure associated with a reaction or decay.

Its Architectural Magnitude is

$$\Pi_{\text{process}}.$$

Unlike the particle magnitude,

$$\Pi_{\text{process}}$$

characterizes the architecture of the transition itself.

### Discussion

This distinction is fundamental.

Particle architectures determine intrinsic structure.

Process architectures describe structural transformations.

The two quantities should therefore not be confused.

## 7.2 Architectural Classification of Reactions

Every perturbative reaction

$$i \rightarrow f$$

possesses an associated process graph

$$D_{\text{process}}.$$

Applying the Graph–Architecture Mapping yields

$$D_{\text{process}} \rightarrow (a, b, c, d).$$

Consequently,

every perturbative reaction possesses an Architectural Magnitude

$$\Pi_{\text{process}}(D).$$

This observation immediately defines an architectural classification of reactions independent of the specific interaction.

Example

Tree-level weak decay

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu.$$

The corresponding Feynman graph possesses

- propagator structure,
- one weak interaction vertex,
- no loops,

and therefore a specific architectural coordinate.

The same procedure applies to every perturbative process.

### 7.3 Architectural Complexity

The Architectural Magnitude naturally suggests a measure of structural complexity.

Definition 10 (Architectural Complexity)

The logarithmic quantity

$$\boxed{C(D) = \ln \Pi(D)}$$

is called the **architectural complexity** of the graph.

Using the definition of the Architectural Magnitude,

$$C(D) = a \ln 2 + b \ln 3 + c \ln(3\pi) + \frac{d}{8} \ln A.$$

Remark

Unlike

$\Pi$ ,

the architectural complexity is additive.

Indeed,

$$C(D_1 \sqcup D_2) = C(D_1) + C(D_2).$$

Thus

$C$

behaves analogously to entropy-like quantities.

**Theorem 16 (Additivity of Architectural Complexity)**

For disjoint graph unions,

$$\boxed{C(D_1 \sqcup D_2) = C(D_1) + C(D_2).}$$

Proof

Immediate from

$$C = \ln \Pi$$

together with the multiplicativity established in Theorem 9.

■

This theorem is stronger than the one about  $\Pi$

Notice that

$$C = \ln \Pi$$

is mathematically simpler than

$$\Pi.$$

The additive structure is likely to be more useful than the multiplicative one.

## 7.4 Possible Relation to Transition Rates

The Architectural Magnitude suggests the possibility that reaction complexity influences transition probabilities.

We therefore introduce the following hypothesis.

Hypothesis 1

The transition probability of a perturbative process may depend on both

- its dynamical amplitude,

and

- its architectural complexity.

Symbolically,

$$\Gamma = F(C, \mathcal{M}, Q),$$

where

$$F$$

is presently unknown.

Discussion

This formulation differs significantly from earlier GFAT versions.

We **do not** propose

$$\Gamma = \Pi^2 \rho(Q).$$

Instead,

we merely state that

architectural complexity

may be one independent variable governing transition rates.

Determining the function

$$F$$

becomes an experimental and mathematical problem.

## 7.5 First Exploratory Comparisons

Representative decay processes indicate that reactions with similar perturbative graph structures often possess similar architectural coordinates.

Whether this correlation extends quantitatively to decay widths remains an open question.

Consequently,

the present work interprets the Architectural Magnitude primarily as

a structural invariant,

not yet as a predictive decay formula.

## 7.6 Open Problem

The central physical problem may therefore be formulated as follows.

Problem 2

Determine whether there exists a universal functional relationship

$$\boxed{\Gamma = F(\Pi, Q, g_i, m_i)}$$

or equivalently

$$\boxed{\Gamma = F(C, Q, g_i, m_i)}$$

valid across perturbative quantum field theory.

## 8. Discussion

The primary objective of the present work has been to introduce a new invariant associated with perturbative Feynman graphs. Unlike perturbative amplitudes, which depend on the underlying quantum field theory, the proposed Architectural Magnitude depends exclusively on graph architecture.

Consequently, the theory should be regarded primarily as a contribution to the mathematical study of perturbative graph structures.

### 8.1 A New Invariant for Feynman Graphs

The history of perturbative quantum field theory has repeatedly demonstrated that graph-theoretical invariants provide important insight into physical processes.

Examples include

- the Euler characteristic,
- the first Betti number,
- Kirchhoff graph polynomials,
- Symanzik polynomials,
- the Connes–Kreimer Hopf algebra.

Each of these constructions emphasizes one particular aspect of graph structure.

The Architectural Magnitude differs in that it combines four structural sectors into one multiplicative invariant,

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}.$$

Its principal novelty lies not in replacing existing invariants, but in integrating several architectural aspects into a single quantity.

### 8.2 Architectural Coordinates

The introduction of the coordinate system

$$(a \cdot b \cdot c \cdot d)$$

provides a geometric interpretation of perturbative graph space.

Instead of studying graphs individually,

one may investigate subsets of

$$\mathbb{N}_0^4.$$

Questions immediately arise.

Which coordinate tuples are realizable?

Which correspond to tree graphs?

Which correspond to primitive divergent graphs?

Which correspond to renormalizable subgraphs?

These questions suggest a new geometric viewpoint on perturbative quantum field theory.

### 8.3 Relation to Existing Mathematical Structures

The Architectural Magnitude should be viewed as complementary to existing graph invariants.

It does not replace

- graph homology,
- graph polynomials,
- Hopf algebras,
- graph spectra,

but rather introduces an additional invariant describing architectural organization.

Whether these invariants are independent,

or whether one can be reconstructed from the others,

remains unknown.

### 8.4 Mathematical Problems

The present work leaves several important mathematical questions unresolved.

Problem 1

Provide an intrinsic graph-theoretical definition of the recursion coordinate

$$R(D).$$

At present,

$R$  depends on the chosen GFAT construction.

A canonical graph-theoretical definition would considerably strengthen the theory.

## Problem 2

Characterize the admissible region of

$$\mathbb{N}_0^4.$$

Not every tuple

$$(a, b, c, d)$$

may correspond to a realizable perturbative graph.

A complete characterization would amount to an existence theorem for architectural coordinates.

## Problem 3

Determine whether the Architectural Magnitude satisfies a universal characterization theorem.

Instead of defining

$$\Pi,$$

one would like to prove

that it is the unique invariant satisfying an appropriate collection of axioms.

Such a theorem would substantially strengthen the mathematical foundations of GFAT.

## Problem 4

Determine whether

is related to

- graph entropy,
- graph energy,
- graph complexity,
- graph spectra,

or other established graph invariants.

### Problem 5

Investigate whether the coordinate system

$$(a, b, c, d)$$

extends naturally to

- lattice gauge theory,
- tensor networks,
- spin networks,
- non-perturbative Green functions.

## 8.5 Future Perspective

The present work represents only a first step toward an invariant theory of perturbative Feynman graphs.

If the Architectural Magnitude proves useful,

it may become one member of a larger family of architectural invariants describing different aspects of graph structure.

Future work should therefore investigate

- canonical normalization,
- universality,
- category-theoretic formulations,
- computational algorithms,
- and applications to realistic perturbative calculations.

## 9. Towards a Characterization Theorem

The previous sections introduced the Architectural Magnitude as a multiplicative invariant on the class of finite perturbative Feynman graphs. A natural mathematical question is whether this invariant is merely one possible construction or whether it is characterized uniquely by a small set of structural principles.

The purpose of this section is to formulate this problem precisely and to identify the mathematical conditions under which a uniqueness theorem may be established.

### 9.1 Axioms for Architectural Invariants

Let

$$\mathfrak{I}: \mathcal{G} \rightarrow \mathbb{R}_{>0}$$

be a positive-valued invariant on the class

$$\mathcal{G}$$

of finite perturbative Feynman graphs.

We require the following axioms.

Axiom A (Graph Isomorphism)

If

$$D_1 \simeq D_2,$$

then

$$\mathfrak{I}(D_1) = \mathfrak{I}(D_2).$$

### Axiom B (Multiplicativity)

For disjoint graph unions,

$$\mathfrak{I}(D_1 \sqcup D_2) = \mathfrak{I}(D_1)\mathfrak{I}(D_2).$$

### Axiom C (Elementary Locality)

Insertion of one elementary architectural operation changes the invariant only through the corresponding architectural sector.

### Axiom D (Architectural Independence)

Propagation,

interaction,

holonomy,

and

recursion contribute independently.

### Theorem 16 (General Representation)

Every invariant satisfying Axioms A–D possesses the form

$$\boxed{\mathfrak{I}(D) = \lambda_P^{a(D)} \lambda_I^{b(D)} \lambda_H^{c(D)} \lambda_R^{d(D)},}$$

for positive constants

$$\lambda_P, \lambda_I, \lambda_H, \lambda_R.$$

### Proof

By multiplicativity and independence, each elementary operation induces a multiplicative action on the invariant. Since the architectural coordinates count the number of primitive operations in each sector, repeated application yields exponential dependence on the corresponding coordinate. The product over the four independent sectors gives the stated representation.

### Corollary 16.1

The exponential form of the Architectural Magnitude is **forced by the axioms**.

Only the normalization constants remain undetermined.

## 9.2 The Canonical Normalization Problem

The remaining mathematical question is whether the constants

$$(\lambda_P, \lambda_I, \lambda_H, \lambda_R)$$

can themselves be characterized by intrinsic structural principles.

Within the current GFAT framework, the canonical choice is

$$\boxed{\lambda_P = 2, \lambda_I = 3, \lambda_H = 3\pi, \lambda_R = A^{1/8}.$$

The first two normalizations are motivated by the combinatorics of the smallest connected transport and branching graphs. The latter two currently arise from architectural principles of holonomy and recursive closure and remain subjects of ongoing investigation.

### Conjecture 1 (Canonical Characterization)

The Architectural Magnitude

$$\Pi(D) = 2^a 3^b (3\pi)^c A^{d/8}$$

is the unique invariant satisfying Axioms A–D together with the canonical normalization principles of GFAT.

### Discussion

A proof of Conjecture 1 would elevate the Architectural Magnitude from a constructed invariant to a **characterized invariant**. In mathematics, such characterization theorems are often more fundamental than explicit formulas because they explain *why* a particular object is canonical.

Establishing this conjecture requires deeper analysis of the architectural bases, particularly the derivation of the holonomy and recursion normalizations from intrinsic geometric or graph-theoretical principles.

## 10. Canonical Operations on Feynman Graphs

The preceding sections introduced the Architectural Magnitude as a multiplicative invariant associated with perturbative Feynman graphs. A natural question is whether this invariant behaves canonically under elementary graph transformations.

The purpose of this section is to investigate the action of elementary graph operations on the architectural coordinates and on the corresponding Architectural Magnitude.

### 10.1 Elementary Graph Operations

Perturbative Feynman graphs are constructed recursively from elementary graphical operations. In the present framework we distinguish four primitive operations corresponding to the four architectural sectors.

Definition 11 (Primitive Architectural Operations)

Let  $D$  be a finite perturbative Feynman graph.

The following elementary operations are defined.

#### Propagation insertion

Insertion of one propagator.

Symbolically,

$$T_P: D \rightarrow D'.$$

#### Interaction insertion

Insertion of one interaction vertex.

$$T_I: D \rightarrow D'.$$

### Holonomy insertion

Insertion of one independent loop.

$$T_H: D \rightarrow D'.$$

### Recursive insertion

Insertion of one recursive subgraph.

$$T_R: D \rightarrow D'.$$

Each primitive operation modifies exactly one architectural coordinate.

### Theorem 17 (Coordinate Transformation Law)

Under the elementary operations the architectural coordinates satisfy

$$\begin{aligned} T_P: (a, b, c, d) &\mapsto (a + 1, b, c, d), \\ T_I: (a, b, c, d) &\mapsto (a, b + 1, c, d), \\ T_H: (a, b, c, d) &\mapsto (a, b, c + 1, d), \\ T_R: (a, b, c, d) &\mapsto (a, b, c, d + 1). \end{aligned}$$

Proof

Immediate from the definitions of the architectural coordinates.

■

### Corollary 17.1

The four primitive operations generate translations on the architectural lattice

$$\mathbb{N}_0^4.$$

## 10.2 Transformation of the Architectural Magnitude

Since

$$\Pi = \lambda_p^a \lambda_l^b \lambda_H^c \lambda_R^d,$$

Theorem 17 immediately implies transformation rules for the Architectural Magnitude.

Theorem 18 (Transformation Law)

The elementary operations satisfy

$$\begin{aligned} T_p: \Pi &\mapsto \lambda_p \Pi, \\ T_l: \Pi &\mapsto \lambda_l \Pi, \\ T_H: \Pi &\mapsto \lambda_H \Pi, \\ T_R: \Pi &\mapsto \lambda_R \Pi. \end{aligned}$$

For the canonical GFAT normalization,

$$\begin{aligned} T_p: \Pi &\mapsto 2\Pi, \\ T_l: \Pi &\mapsto 3\Pi, \\ T_H: \Pi &\mapsto 3\pi \Pi, \\ T_R: \Pi &\mapsto A^{1/8}\Pi. \end{aligned}$$

Proof

Each elementary operation increases exactly one architectural coordinate by one while leaving the remaining coordinates unchanged. The result follows directly from the exponential representation established in Theorem 16.

■

## 10.3 Architectural Transformation Group

The elementary operations naturally generate an algebraic structure.

Definition 12

Let

$$\mathcal{T} = \langle T_P, T_I, T_H, T_R \rangle$$

denote the set generated by repeated application of the primitive architectural operations.

Proposition 1

Assuming that the elementary operations are independent,

the transformations commute,

$$T_i T_j = T_j T_i.$$

Consequently,

$$\mathcal{T}$$

forms a free commutative monoid.

Proof

Each primitive operation modifies a different architectural coordinate. Coordinate increments therefore commute under composition.

■

## 10.4 Architectural Lattice

The coordinate system

$$(a, b, c, d)$$

induces a lattice structure on the set of architectural classes.

Each elementary operation corresponds to a unit translation along one coordinate axis.

Accordingly,

the Architectural Magnitude is naturally interpreted as a multiplicative weight function on the architectural lattice.

## **Declarations**

### **Author Contributions**

The author conceived the theoretical framework, developed the mathematical formalism, derived the graph-theoretical constructions, performed the mathematical analysis, interpreted the results, and wrote the manuscript.

### **Funding**

This research received no external funding.

### **Institutional Review Board Statement**

Not applicable.

### **Informed Consent Statement**

Not applicable.

### **Data Availability Statement**

No new experimental data were generated or analyzed during this study. All mathematical derivations are contained within the manuscript.

### **Acknowledgements**

The author thanks colleagues and members of the theoretical physics community whose work on Green functions, graph theory, quantum field theory, and mathematical physics has provided the mathematical foundations upon which the present work builds.

### Conflicts of Interest

The author declares no competing interests.

### References

1. Green G. *An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism*. Nottingham: T Wheelhouse; 1828.
2. Courant R, Hilbert D. *Methods of Mathematical Physics*. Vol. I. New York: Wiley; 1953.
3. Morse PM, Feshbach H. *Methods of Theoretical Physics*. New York: McGraw-Hill; 1953.
4. Reed M, Simon B. *Methods of Modern Mathematical Physics*. Vol. II. New York: Academic Press; 1975.
5. Arfken GB, Weber HJ, Harris FE. *Mathematical Methods for Physicists*. 7th ed. Amsterdam: Academic Press; 2013.
6. Peskin ME, Schroeder DV. *An Introduction to Quantum Field Theory*. Reading (MA): Addison-Wesley; 1995.
7. Weinberg S. *The Quantum Theory of Fields*. Vol. I. Cambridge: Cambridge University Press; 1995.
8. Itzykson C, Zuber JB. *Quantum Field Theory*. New York: McGraw-Hill; 1980.
9. Ryder LH. *Quantum Field Theory*. 2nd ed. Cambridge: Cambridge University Press; 1996.
10. Bogoliubov NN, Shirkov DV. *Introduction to the Theory of Quantized Fields*. 3rd ed. New York: Wiley; 1980.
11. Feynman RP. Space-time approach to quantum electrodynamics. *Phys Rev*. 1949;76:769-789.
12. Dyson FJ. The S matrix in quantum electrodynamics. *Phys Rev*. 1949;75:1736-1755.
13. Dyson FJ. The radiation theories of Tomonaga, Schwinger, and Feynman. *Phys Rev*. 1949;75:486-502.
14. Wick GP. The evaluation of the collision matrix. *Phys Rev*. 1950;80:268-272.
15. Schwinger J. On the Green's functions of quantized fields I. *Proc Natl Acad Sci USA*. 1951;37:452-455.
16. Yang CN, Mills RL. Conservation of isotopic spin and isotopic gauge invariance. *Phys Rev*. 1954;96:191-195.
17. 't Hooft G, Veltman M. Regularization and renormalization of gauge fields. *Nucl Phys B*. 1972;44:189-213.

18. Gross DJ, Wilczek F. Ultraviolet behavior of non-Abelian gauge theories. *Phys Rev Lett.* 1973;30:1343-1346.
19. Politzer HD. Reliable perturbative results for strong interactions. *Phys Rev Lett.* 1973;30:1346-1349.
20. Wilson KG. Confinement of quarks. *Phys Rev D.* 1974;10:2445-2459.
21. Wilson KG. The renormalization group and critical phenomena. *Rev Mod Phys.* 1975;47:773-840.
22. Collins JC. *Renormalization*. Cambridge: Cambridge University Press; 1984.
23. Connes A, Kreimer D. Hopf algebras, renormalization and noncommutative geometry. *Commun Math Phys.* 1998;199:203-242.
24. Kreimer D. On the Hopf algebra structure of perturbative quantum field theories. *Adv Theor Math Phys.* 1998;2:303-334.
25. Bollobás B. *Modern Graph Theory*. New York: Springer; 1998.
26. Diestel R. *Graph Theory*. 5th ed. Berlin: Springer; 2017.
27. Tutte WT. *Graph Theory*. Reading (MA): Addison-Wesley; 1984.
28. Hatcher A. *Algebraic Topology*. Cambridge: Cambridge University Press; 2002.
29. Witten E. Quantum field theory and the Jones polynomial. *Commun Math Phys.* 1989;121:351-399.
30. Nakanishi N. *Graph Theory and Feynman Integrals*. Philadelphia: Gordon and Breach; 1971.